

Theory On React:

- Identify the differences between React components and Javascript functions.

React Components	JavaScript Functions
Returns JSX(UI elements)	Returns any JavaScript value
Used to build user interface	Used for general programming tasks
Can use React features like props, states and hooks.	Cannot directly use React features
Component names must begin with capital letter	Function name can start with valid identifier
Render by React	Executed directly by JavaScript

- Identify the types of components.
 1. Functional Components
 2. Class Components
- Explain Class Component.

A Class Component is created using an ES6 class that extends `React.Component`. It can maintain its own state, use lifecycle methods, and must include a `render()` method that returns JSX.

Example:

```
import React, { Component } from 'react';

class Welcome extends Component {
  render() {
    return <h1>Welcome to React</h1>;
  }
}

export default Welcome;
```

- Explain Functional Components

A Functional Component is a JavaScript function that accepts props as input and returns JSX. It is simpler, easier to understand, and is the recommended approach in modern React because it supports Hooks.

Example:

```
function Welcome() {  
  return <h1>Welcome to React</h1>;  
}  
  
export default Welcome;
```

- Define Component Constructor.

The constructor is a special method used only in Class Components. It is automatically called when a component is created and is mainly used to initialize the component's state and bind event handlers.

Example:

```
class Welcome extends React.Component {  
  constructor(props) {  
    super(props);  
    this.state = {  
      message: "Hello"  
    };  
  }  
  
  render() {  
    return <h1>{this.state.message}</h1>;  
  }  
}
```

- Define render() function.

The render() function is a mandatory method in a Class Component. It specifies what should be displayed on the screen by returning JSX. Whenever the component's state or props change, React automatically calls the render() method to update the user interface.

```
render() {  
  return (  
    <h1>Welcome to React</h1>  
  );  
}
```

HandsOn Practice:

```
JS App.js myfirstreact\... M JS Home.js U JS App.js studentapp\... M X JS About.js U JS Contact.js U
1 import logo from './logo.svg';
2 import './App.css';
3 import Home from './Components/Home';
4 import About from './Components/About';
5 import Contact from './Components/Contact';
6
7
8 function App() {
9   return (
10     <div className='container'>
11       <Home/>
12       <About/>
13       <Contact/>
14     </div>
15   );
16 }
17
18
19 export default App;
20
```

```
View Go Run Terminal Help React_HandsOn
... JS App.js myfirstreact\... M X JS Home.js U JS App.js studentapp\... M JS About.js U X JS Contact.js U
1 import React from "react";
2
3 function About() {
4   return (
5     <div>
6       <h2>About Page</h2>
7     </div>
8   );
9 }
10
11 export default About;
```

```
JS App.js myfirstreact\... M JS Home.js U JS App.js studentapp\... M JS About.js U JS Contact.js U X
1 import React from "react";
2
3 function Contact() {
4   return (
5     <div>
6       <h2>Contact Page</h2>
7     </div>
8   );
9 }
10
11 export default Contact;
```

```
JS App.js myfirstreact\... M JS Home.js U JS App.js studentapp\... M JS About.js U # App.css JS Contact.js U X
1 import React from "react";
2
3 function Contact() {
4   return (
5     <div>
6       <h2>Contact Page</h2>
7     </div>
8   );
9 }
10
11 export default Contact;
```

```
JS App.js myfirstreact\... M JS Home.js U JS App.js studentapp\... M X JS About.js U # App.css JS Contact.js U
1 import logo from './logo.svg';
2 import './App.css';
3 import Home from './Components/Home';
4 import About from './Components/About';
5 import Contact from './Components/Contact';
6
7
8 function App() {
9   return (
10     <div className='container'>
11       <Home/>
12       <About/>
13       <Contact/>
14     </div>
15   );
16 }
17
18
19 export default App;
20
```

